Part A — Armington with a productivity spillover

We add to the model of Topic 6 a production spillover, $A_i = \bar{A}_i L_i^{\lambda}$ with $\lambda \geq 0$ and $\bar{A}_i$ exogenous; nothing else changes. In the code the spillover is `par.lambda`, read by `arm_solve.m` (line `Ae = A .* L.^lam`) and by `arm_invert.m` (line `A = Aeff ./ L.^lam`).

(a) Equilibrium conditions and identification

Prices. Location i produces one variety with labour only, under constant returns and perfect competition, so the price at origin is w_i/A_i and the delivered price in n is $p_{ni} = \tau_{ni} w_i/A_i$ (Topic 6, slide 19). With the spillover,

$$p_{ni} = \frac{\tau_{ni}\, w_i}{\bar{A}_i L_i^{\lambda}}. \tag{1}$$

Price index. With CES demand and $\phi_{ni} = \tau_{ni}^{1-\sigma}$ (slide 22),

$$P_n^{1-\sigma} = \sum_i \phi_{ni} \left(\frac{w_i}{\bar{A}_i L_i^{\lambda}} \right)^{1-\sigma} = \sum_i \phi_{ni}\, \bar{A}_i^{\sigma-1} L_i^{\lambda(\sigma-1)} w_i^{1-\sigma}. \tag{2}$$

Trade shares. The share of n's spending on goods from i (slide 21) is

$$\pi_{ni} = \frac{\phi_{ni} \left(w_i/\bar{A}_i L_i^{\lambda} \right)^{1-\sigma}}{\sum_k \phi_{nk} \left(w_k/\bar{A}_k L_k^{\lambda} \right)^{1-\sigma}}. \tag{3}$$

Trade balance. With zero profits the wage bill of i equals the sales of its variety, $w_i L_i = \sum_n \pi_{ni} X_n$ (slide 26). Rents are redistributed to residents, so income per resident is w_i/μ and spending on goods is $X_n = \mu\,(w_n/\mu) L_n = w_n L_n$ (slide 27). Hence

$$w_i L_i = \sum_n \frac{\phi_{ni} \left(w_i/\bar{A}_i L_i^{\lambda} \right)^{1-\sigma}}{\sum_k \phi_{nk} \left(w_k/\bar{A}_k L_k^{\lambda} \right)^{1-\sigma}}\; w_n L_n. \tag{4}$$

Population now appears on both sides: on the left as the labour force, on the right through productivity, with exponent $\lambda(\sigma - 1)$.

$\bar{A}_i$ and λ enter only through A_i. In (2)–(4) $\bar{A}_i$ and λ never appear separately, only in the product $\bar{A}_i L_i^{\lambda} = A_i$. The remaining conditions do not contain productivity except through P_n and w_n: rents $r_i = \frac{1-\mu}{\mu} \frac{w_i L_i}{H_i}$, real utility $W_i = B_i\,(w_i/\mu)/(P_i^{\mu} r_i^{1-\mu})$ (slide 28) and the Fréchet shares $L_i/\bar{L} = W_i^{\varepsilon} / \sum_s W_s^{\varepsilon}$ (slide 29). So the whole equilibrium depends on $(\bar{A}, \lambda)$ only through A: two parametrizations with the same $\bar{A}_i L_i^{\lambda}$ in every location are the same economy.

Inversion. Given data (L_i, w_i, H_i), the matrix τ and σ, the inversion of Topic 6 (slides 42–43) solves the trade balance (4) for the competitiveness $c_i = (w_i/A_i)^{1-\sigma}$ by a fixed point, and reads $A_i = w_i\, c_i^{1/(1-\sigma)}$. Since trade shares depend only on ratios of c_i, A is recovered up to scale, fixed by a geometric-mean normalization. This step uses L and w as data and never uses λ: it recovers the composite A_i exactly (`inv.Aeff`). The fundamental needs λ:

$$\bar{A}_i = A_i\, L_i^{-\lambda}, \tag{5}$$

so it is recovered only for a given λ (`inv.A`).

Every λ fits the data equally well. For any $\lambda' \geq 0$ set $\bar{A}_i(\lambda') = A_i L_i^{-\lambda'}$. Then $\bar{A}_i(\lambda') L_i^{\lambda'} = A_i$ in every location, so prices, trade shares and the trade balance are those of the true economy, and (L, w) are reproduced exactly. One cross-section gives N equations that pin down N composites; splitting them into N fundamentals and one λ requires information the cross-section does not contain. As in Part C of Homework 1, the data identify A, not $(\bar{A}, \lambda)$.

(b) The same data inverted with three values of λ

What we did. In `ARM1_geography.m` we added a section after act 6. The data are the baseline equilibrium of act 5, generated with $\lambda = 0$ (`par` has no `lambda` field). We invert them three times, changing λ only in a copy of the parameters, so that the data-generating block at the top is untouched:

```
p = par;  p.lambda = lam_list(k);          % lam_list = [0 0.07 0.25]
inv = arm_invert(data, p, tau_l, Ltot);
errAeff   = max(abs(log(inv.Aeff ./ Atrue)));
sdAbar(k) = std(log(inv.A));
```

λ	$\max_i \lvert \log(A_i^{\text{inv}}/A_i^{\text{true}}) \rvert$	$\text{sd}(\log \bar{A}_i)$
0	1.17×10^{-12}	0.3033
0.07	1.17×10^{-12}	0.2715
0.25	1.17×10^{-12}	0.2041

Table 1: Same data (L, w, H), three values of λ in the inversion. The fixed point converges in 4346 iterations in all three cases; $\text{corr}(\log A, \log L) = 0.814$, $\text{sd}(\log L) = 0.575$.

Why the error on A_i does not depend on λ. The composite comes out of the fixed point on the trade balance, which uses only L, w and τ: `lam` enters `arm_invert` only after the loop, in `A = Aeff ./ L.^lam`. So `inv.Aeff` is the same vector for every λ — same error, 1.17×10^{-12}, and even the same number of iterations, 4346 — and matches the true productivity up to numerical precision (the data were generated with $\lambda = 0$, so the true composite equals `Atrue`). This is point (a) in numbers: the composite is identified.

Why the dispersion of $\log \bar{A}_i$ does. The fundamental is $\log \bar{A}_i = \log A_i - \lambda \log L_i$, so

$$\text{var}(\log \bar{A}) = \text{var}(\log A) - 2\lambda \,\text{cov}(\log A, \log L) + \lambda^2 \,\text{var}(\log L). \tag{6}$$

Productive locations attract people, so $\log A$ and $\log L$ are strongly positively correlated (0.814 here): the larger λ, the more of the observed productivity is attributed to size and the less to fundamentals, and the fundamentals look more alike. With $\text{sd}(\log A) = 0.3033$, $\text{sd}(\log L) = 0.575$ and correlation 0.814, the formula reproduces the table exactly: 0.2715 at $\lambda = 0.07$ and 0.2041 at $\lambda = 0.25$. The variance falls as long as λ is below $\text{cov}(\log A, \log L)/\,\text{var}(\log L) = 0.814 \times 0.3033/0.575 \approx 0.43$, the OLS slope of $\log A$ on $\log L$. Same data, same fit, different story about where productivity comes from.

(c) The +20% shock with three values of λ

What we did. We repeat act 5 — the same fundamentals `fund3` and the same +20% shock to $\bar{A}_j$ in location $j = 8$, the middle of the line — with `p` in place of `par` in both calls to `arm_solve`. The baseline equilibrium also changes with λ, because $A_i = \bar{A}_i L_i^{\lambda}$.

λ	ΔL_j	ΔL_{j-1}	ΔL_{j+1}	ΔA_j (effective)	$\Delta L_j/\Delta L_j(\lambda{=}0)$
0	+26.91%	−1.02%	−0.28%	+20.00%	1.00
0.07	+30.34%	−1.01%	+0.03%	+22.25%	1.13
0.25	+45.28%	−0.69%	+2.46%	+31.74%	1.68

Table 2: Population change in the shocked location and its two neighbours, and change in its effective productivity $\bar{A}_j L_j^{\lambda}$.

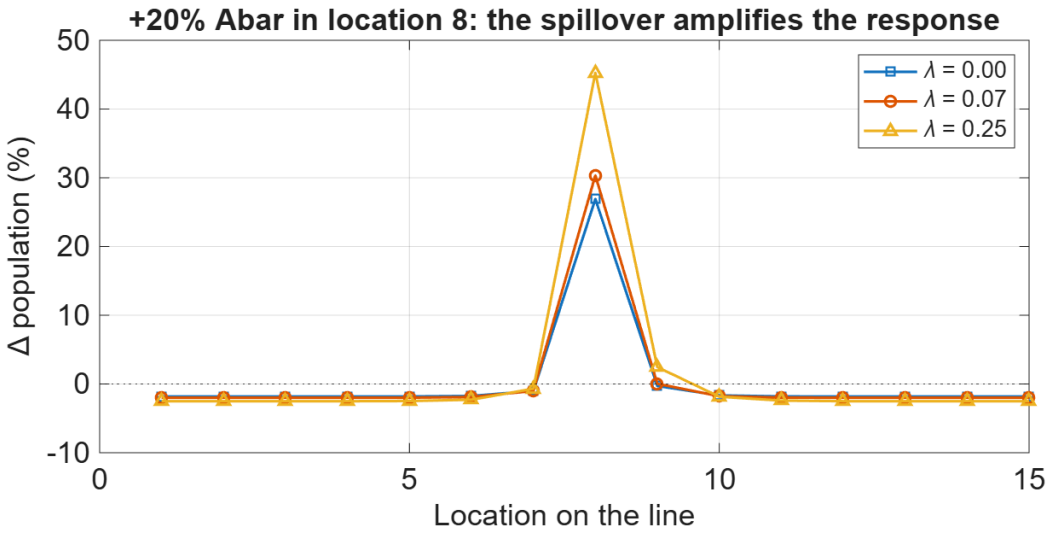


Figure 1: Population change along the line after +20% $\bar{A}_j$, for $\lambda = 0$, 0.07 and 0.25.

Why the response grows with λ. The endogenous variable that feeds back into productivity is the population of the shocked location, L_j, and it moves through the labour market: workers migrate towards the higher real wage. With $\lambda > 0$ they bring productivity with them, since $A_j = \bar{A}_j L_j^{\lambda}$. The goods market closes the loop: higher A_j lowers the delivered price of j's variety everywhere, its sales and wage bill rise, utility in j rises further and attracts more workers. The last column of the table measures this amplification. The effective productivity obeys an exact identity,

$$1 + \Delta A_j = 1.20\,(1 + \Delta L_j)^{\lambda}, \tag{7}$$

so the shock to $\bar{A}_j$ is 20% in every case, but the productivity the economy actually sees rises by 22.2% at $\lambda = 0.07$ ($1.20 \times 1.3034^{0.07} = 1.2225$) and by 31.7% at $\lambda = 0.25$ ($1.20 \times 1.4528^{0.25} = 1.3174$). The population response is amplified by a factor 1.13 and 1.68 relative to $\lambda = 0$; with $\lambda = 0$ we recover exactly act 5 (+26.906%).

The neighbours. Two forces act on them (slide 39). Migration pulls workers towards j; but the neighbours buy a large share of j's variety, now cheaper, so their price index falls and their real wage rises ("proximity protects"). With $\lambda > 0$ j's good becomes cheaper still, so the second force is stronger. The table shows it: the left neighbour's loss shrinks from −1.02% to −0.69%, and the right neighbour goes from −0.28% to +0.03% and +2.46%, a gain. The two neighbours differ because their own fundamentals differ (they are drawn at random in act 5). Far from j the price-index gain is small and only the migration pull remains: the farthest locations lose −1.83% at $\lambda = 0$ (act 5) and about −2.4% at $\lambda = 0.25$ (figure). The spillover makes the shock larger and more concentrated in space: the booming location and its neighbours gain more, remote places lose more.

Why it matters for policy. By point (b) the three values of λ fit the same data equally well, yet the same local productivity gain — a new plant, a subsidy, an infrastructure — raises the population of the location by 26.9% with $\lambda = 0$ and by 45.3% with $\lambda = 0.25$. The policy answer depends on a parameter the cross-section cannot identify.